

Laboratory 6. Orbits of planar systems

I. Hyperbolic equilibria.

1. We consider the planar nonlinear system $\dot{x} = 2x - x^2 - xy, \quad \dot{y} = -y + xy$.

a) Find all its equilibrium points.

b) For each equilibrium point, denoted $\eta^* \in \mathbb{R}^2$,

bi) compute the Jacobian matrix $A = Jf(\eta^*)$, where $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is the function that defines the given system, i.e. $f_1(x, y) = 2x - x^2 - xy$ and $f_2(x, y) = -y + xy$.

bii) Find the eigenvalues of the matrix $A = Jf(\eta^*)$. Notice that there is no eigenvalue with 0 real part, which means that each equilibrium point is hyperbolic.

biii) Write explicitly the linear system $\dot{X} = AX$ and specify its type and stability.

biv) Specify the stability of the equilibrium point η^* of the given nonlinear system.

bv) Find the solution of the IVP $\dot{X} = AX, X(0) = \eta$, for all $\eta \in \{(1, 0), (1, 1), (0, 1), (-1, 1), (-1, 0), (-1, -1), (0, -1), (1, -1)\}$.

Plot the corresponding orbits in the same plane.

c) Represent the direction field of the system in the box $[-1, 2.2] \times [-1, 1.2]$ of the phase plane. (In Maple use **dfieldplot**.) Notice that this box contains all the equilibrium points. Localize them. Now focus the image on smaller boxes around each equilibrium, like $[-0.2, 0.2] \times [-0.2, 0.2]$, $[1.2, 2.2] \times [-0.2, 0.2]$ and, respectively, $[0.2, 1.2] \times [0.2, 1.2]$. We will show in a lecture that the direction field is tangent to the orbits. Can you identify the shape of the orbits around each equilibrium point?

d) Using **DEplot** in Maple, plot few orbits near each equilibrium point. For better results, first plot simultaneously only orbits near the same equilibrium point. For the variable time t take, for example, the interval $[0, 1]$ for various reasons: to identify better on the picture the initial and, respectively the future states (that is why we take $t \geq 0$) and, on the other hand, the interval is sufficiently small to not accumulate errors. Try also on larger intervals, like $[0, 20]$, sometimes it works. Choose the initial states to obtain a nice picture.

e) Gluing all the information obtained until now, sketch the phase portrait of this system in your notebook. Also, write the equilibria and the corresponding stability character.

2. This exercise has the same requirements as the previous one for the planar system $\dot{x} = x - 2xy, \quad \dot{y} = x^2/2 - y$. Of course, at c) choose suitable boxes for this system.

II. Non-hyperbolic equilibria.

3. We consider the conservative (or undamped) pendulum system

$$\dot{x} = y, \quad \dot{y} = -4 \sin x$$

and the function

$$H : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad H(x, y) = \frac{1}{2}y^2 - 4 \cos x,$$

which is the total mechanical energy of the pendulum.

a) Notice that $(0, 0)$ is an equilibrium point and justify that it is not hyperbolic.

b) Notice that $(0, 0)$ is a critical point of H . Plot the level curves of H in the box $[-5, 5] \times [-5, 5]$. Notice that the level curves which are close to the critical point $(0, 0)$ are closed curves. If you want to convince yourself of this, represent the level curves of H in smaller boxes.

c) Check that

$$\frac{\partial H}{\partial x} * y - 4 \frac{\partial H}{\partial y} * \sin x = 0 \quad \text{in } \mathbb{R}^2.$$

From here, we deduce that H is a first integral of the pendulum system. This implies that the orbits of the pendulum system lie on the level curves of H .

d) Justify that the oscillations of the pendulum with small initial data are periodic.

4. We consider the Lotka-Volterra system (also called the predator-prey system; let's say that the foxes are the predators and the rabbits are the preys)

$$\dot{x} = x - xy, \quad \dot{y} = -0.3y + 0.3xy.$$

Moreover, we have that $100 * x(t)$ is the number of rabbits at time t , while $100 * y(t)$ is the number of foxes at time t in a given common habitat. So, the foxes eat rabbits and only rabbits, while the rabbits eat carrots. There are enough carrots, thus the rabbits can not die of starvation. We also assume that the borders of the habitat are closed, thus no one can enter or go out.

a) Note that $(1, 1)$ is an equilibrium point and show that it is non-hyperbolic.

b) Let $H : (0, \infty) \times (0, \infty) \rightarrow \mathbb{R}$, $H(x, y) = y - \ln y + 0.3(x - \ln x)$. Check that H is a first integral of the Lotka-Volterra system in the region $(0, \infty) \times (0, \infty)$, i.e. check that

$$\frac{\partial H}{\partial x} * (x - xy) + \frac{\partial H}{\partial y} * (-0.3y + 0.3xy) = 0.$$

c) Note that $(1, 1)$ is a critical point of H . Plot the level curves of H .

d) Justify that the orbits near $(1, 1)$ are periodic orbits.

e) Assume that at time $t = 0$ there are 100 rabbits and 50 foxes. Describe the dynamics reading the phase portrait by detecting four crucial moments: - when one of the populations has a critical moment (e.g. before it increased, while after this critical moment, it will decrease). This behavior seems realistic to you? Try an explanation.